

Exam Date & Time: 08-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. EXAMINATIONS - JANUARY 2024

SUBJECT: ECE 2125- ELECTROMAGNETIC WAVES

ELECTROMAGNETIC WAVES [ECE 2125]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

Smith Chart may be used for calculations

- 1A) Given $\mathbf{A} = y\mathbf{a}_x + x\mathbf{a}_y + (x^2/(x^2+y^2)^{1/2})\mathbf{a}_z$, transform $\mathbf{A}$ from cartesian to cylindrical coordinate system. Also, find $\mathbf{A} \cdot \mathbf{B}$ if $\mathbf{B} = x\mathbf{a}_x - y\mathbf{a}_y$. (4)
- 1B) A vector of magnitude 10 is directed from $(6, 3\pi/4, 1)$ in cylindrical coordinates toward the origin. Express the vector in Cartesian coordinates. (3)
- 1C) Show that if the electric flux density $\mathbf{D} = (100/\rho)\mathbf{a}_\rho + 40\mathbf{a}_z$ C/m² in a certain region then there are no free charges in that region. (3)
- 2A) Given that the electric flux density $\mathbf{D} = 30e^{-\rho}\mathbf{a}_\rho - 2z\mathbf{a}_z$, evaluate both sides of the divergence theorem for the volume enclosed by $\rho = 2$, $z = 0$, and $z = 5$, and $0 \leq \phi \leq 2\pi$ (4)
- 2B) A point charge at $(1, -1, 3)$ m of charge $300\mu\text{C}$ experiences a force of $\mathbf{F} = 8\mathbf{a}_x - 8\mathbf{a}_y + 4\mathbf{a}_z$ N due to a charge Q at $(3, -3, -2)$ m. Determine Q . (3)
- 2C) Two uniform line charges of density $\rho_L = 4\text{ nC/m}$ lie in the $x = 0$ plane at $y = +1\text{ m}$ and $y = -1\text{ m}$. Find the electric field $\mathbf{E}$ at $(4, 0, 10)$ m. (3)
- 3A) Consider the portion of a sphere given by $r = 4\text{ m}$, $0 < \theta < 0.1\pi$ and $0 < \phi < 0.3\pi$. If the magnetic field is $\mathbf{H} = 6r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$ A/m, evaluate both the sides of Stoke's theorem. (4)
- 3B) Find the magnetic flux crossing the portion of a plane $\phi = \pi/4$, $1\text{ cm} < \rho < 4\text{ cm}$ and $0 < z < 3\text{ m}$, due to a current filament along z -axis carrying 0.5 A current in $\mathbf{a}_z$ direction. (3)
- 3C) The distribution of current across the cross section of a solid cylindrical conductor of radius "a" is given by $\mathbf{J} = \rho^2/a^2$. Determine the magnetic fields inside and outside the conductor. (3)
- 4A) A 1 MHz plane wave is propagating in fresh water having $\mu_r = 1$, $\epsilon_r = 81$ and $\epsilon'' = 0$ in $\mathbf{a}_z$ direction. Determine (i) phase constant β , (ii) wavelength, (iii) velocity, (iv) impedance of the medium, and (v) expressions of electric and magnetic fields. Assume the amplitude $E_{x0} = 1$. (4)
- 4B) Consider an incident wave of frequency 1 MHz in a medium having conductivity 4 S/m , $\epsilon_r = 81$ and $\mu_r = 1$. Determine the skin depth, wavelength and velocity. (3)
- 4C) Given the magnetic field $\mathbf{H} = 300\cos(3 \times 10^8 t - y)\mathbf{a}_z$ A/m in free space, determine the emf around the closed path having the corners at $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$ and $(0, 1, 0)$. (3)
- 5A) A lossless transmission line is terminated in a load of $50\ \Omega$. If the frequency of signal carried by this line is 1 MHz and the line parameters are (4)

$L = 50 \text{ nH/m}$ and $C = 5 \text{ pF/m}$, determine β , λ , R_o , and the VSWR

- 5B) Derive expressions for the circular arcs which comprise the Smith Chart. (3)
- 5C) Starting from an infinitesimal section of a transmission line, derive the transmission line equations and discuss their solutions (3)

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